higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

(8080175)

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09:00–12:00

Drawing instruments and nonprogrammable calculators may be used.

This question paper consists of 6 pages and a formula sheet of 5 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL ELECTRONICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a new page.
5. Use only a black or blue pen.
6. Write neatly and legibly.

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QUESTION 1: AC THEORY

1.1 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (1.1.1–1.1.3) in the ANSWER BOOK.

1.1.1 Selectivity of an RLC network is the ability of a network to respond to a specific frequency and discriminate against all other frequencies.

1.1.2 In parallel networks the current through inductor and capacitor is equal to the supply current.

1.1.3 In a series resonance circuit, any reactance value is the same as the resistance value.

(3 × 1) (3)

1.2 Briefly explain the term, *capacitive coupling*, and state ONE disadvantage of RC coupling. (3)

1.3 Show by means of neatly labelled waveforms, the difference between the *odd harmonics* and *even harmonics*. (4)

1.4 Information on a phasor diagram of a series-parallel circuit is given below:

$V_T = 100 \angle 0° \, V$
$I_T = 0{,}83 \angle -1{,}95° \, A$
$I_A = 0{,}83 \angle 0{,}15° \, A$
$I_B = 0{,}032 \angle -89{,}75° \, A$

1.4.1 Draw a neat-labelled phasor diagram of the circuit. (3)

1.4.2 Calculate the total impedance of the circuit. (2)
[15]

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QUESTION 2: POWER SUPPLIES

2.1 The output of a full wave rectifier with a capacitor filter is further filtered by an RC-filter. The values of R and C are 120 Ω and 150 µF, if a 15 VDC with a 1,5 V ripple is developed across the first capacitor and the line frequency is 50 Hz.

Calculate the following values:

2.1.1 Output VDC with the load resistor of 10 kΩ (3)

2.1.2 The ripple voltage across the load (3)

2.1.3 The percentage ripple factor of the first capacitor (2)

2.2 Voltage multipliers are used as soon as the power supplies require a higher voltage.

Briefly describe the operational principle of a voltage tripler. (4)

2.3 Draw a neat, labelled circuit diagram of a Zener-diode reference source. (3)
[15]

QUESTION 3: AMPLIFIERS

3.1 Explain the term, *stabilisation*. (1)

3.2 A fixed bias voltage amplifier has an input impedance of 2,5 kΩ and output impedance of 3,5 kΩ with a load resistor of 6 kΩ.

Calculate the value of the base resistor with the aid of precise method if the following information about the transistor is known:

$h_{ie} = 13\ k\Omega$; $h_{re} = 2{,}8x10^{-4}$; $R_C = 480\ \Omega$;

$h_{oe} = 30\ siemens$; $h_{fe} = 300$ *and* $R_s = 0$ (6)

3.3 If the input signal is a square waveform of ±2V; 800 Hz and the input resistor is 10 MΩ and feedback capacitor is 0,01µF respectively, accept that the capacitor is discharged originally.

Calculate the output voltage for an integrator operational amplifier. (8)
[15]

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QUESTION 4: IC AND TRANSDUCERS

4.1 Explain the term, *integrated circuit*. (2)

4.2 7812 is a three-terminal voltage regulator and has a three-pin layout. Label the three pins in a 7812 voltage regulator. (3)

4.3 List TWO advantages of 7812 series voltage regulator. (2)

4.4 Draw a neat, labelled block diagram of how transducer can de classified. (3)

4.5 List TWO ways that a Wheatstone bridge thermistor can be used. (2)

4.6 Calculate the number of turns of potentiometer with a percentage resolution of 0,125%. (3)
[15]

QUESTION 5: ELECTRONIC PHASE CONTROL

5.1 List TWO methods on how silicone control rectifiers can be:

5.1.1 Switched ON (2)

5.1.2 Switched OFF (2)

5.2 Draw a neat, labelled circuit diagram and output wave form of a single-phase control circuit which is suitable for a light dimmer in electronics. (6)
[10]

QUESTION 6: MEASURING INSTRUMENTS

6.1 Name FIVE characteristics of a Schmitt trigger converter. (5)

6.2 Name each block diagram of a continuous balance digital voltmeter. (5)
[10]

QUESTION 7: OSCILLATORS

7.1 Draw a neat, labelled circuit diagram of RC-phase shift oscillators. (4)

7.2 Draw a neat, labelled circuit diagram of the pin layout of a 555 precision timer. (4)

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7.3 Complete the following sentences by filling in the missing words. Write only the words next to the question number (7.3.1–7.3.2) in the ANSWER BOOK.

In an a-stable 555 timer, as the voltage is switched on, the (7.3.1) … is discharged and the trigger input is low, which causes the output to be (7.3.2) …. While the discharged transistor is OFF, the capacitor charges up to $^{++}$Vcc through R_A and R_B. (2)

[10]

QUESTION 8

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–E) next to the question number (8.1–8.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
8.1	Application of LCD	A	Transmissive type
8.2	Main types of LCD	B	Reflective type
8.3	A vacuum containing an electron gun that emits beams which can display images.	C	Twisted nematic and in-plane switching
8.4	In this type cells, both glass sheets are transparent so that the light is scattered in the forward direction when the cell becomes active	D	CRT
8.5	In this type cells consist of the reflecting surface of the glass sheet on one end	E	TV sets and computers

(5 × 2) **[10]**

TOTAL: 100

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FORMULA SHEET

$$I = \frac{V}{R} \qquad P = IV = I^2R = \frac{V^2}{R}$$

$$V_T = V_1 + V_2 + V_3 + ... = I_1R_1 + I_2R_2 + I_3R_3 + ...$$

$$I_T = I_1 + I_2 + I_3 + ... = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + ...$$

$$T = RC \qquad T = \frac{L}{R}$$

$$V_R = RC\frac{dv}{dt} \qquad V_C = \frac{1}{RC}\int v_i dt$$

$$X_L = 2\pi fL \qquad X_C = \frac{1}{2\pi fC}$$

$$Z = R + jX_L \qquad Z = R - jX_C$$

$$Z = R + j(X_L - X_C) \qquad I_T = \frac{V_T}{Z_T}$$

$$V_R = I_TR \qquad V_L = I_T(jX_L)$$

$$V_C = I_T(-jX_C) \qquad f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = \frac{V_L}{V_T} = \frac{V_C}{V_T} = \frac{X_L}{R} = \frac{X_C}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{f_r}{f_2 - f_1}$$

$$BW = f_2 - f_1 \qquad \frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$Z_T = \frac{Z_2Z_2}{Z_1 + Z_2} \qquad I_T = I_1 + I_2 = \frac{V}{Z_1} + \frac{V}{Z_2}$$

$$Z_T = \frac{R(jX_L)}{R + jX_L} \qquad \frac{1}{Z_T} = \frac{1}{R} - \frac{j}{X_L}$$

$$I_T = I_R - jI_L \qquad I_T = \frac{V}{R} - j\frac{V}{X_L}$$

$$Z_T = \frac{R(-jX_C)}{R - jX_C} \qquad \frac{1}{Z_T} = \frac{1}{R} + \frac{j}{X_C}$$

$$I_T = I_R + jI_C \qquad I_T = \frac{V}{R} + j\frac{V}{X_C}$$

$$\frac{1}{Z_T} = \frac{1}{R} - j\left(\frac{1}{X_L} - \frac{1}{X_C}\right)$$

$$I_T = I_R - j(I_L - I_C) \qquad I_T = \frac{V}{R} - j\left(\frac{V}{X_L} - \frac{V}{X_C}\right)$$

$$a + jb = \sqrt{a^2 + b^2} \,/\, \underline{\tan^{-1}\frac{b}{a}} = r\,\underline{/\theta}$$

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$$r/\underline{\theta} = r(\cos\theta + j\sin\theta)$$

$$Q = \tan\theta$$

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

$$Z_d = \frac{L}{CR_1}$$

$$V_{rms} = \frac{1}{\sqrt{2}}V_m = 0{,}707\,V_m$$

$$V_{dc} = \frac{2}{\pi}V_m = 0{,}637\,V_m$$

$$\frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{I_S}{I_P}$$

$$V_{dc} = \frac{1}{\pi}V_m = 0{,}318\,V_m$$

$$PIV = V_m$$

$$PIV = 2\,V_m$$

$$R_{r(rms)} = 0{,}385\,V_m$$

$$V_{r(rms)} = 0{,}305\,V_m$$

$$r = \frac{V_{r(rms)}}{V_{dc}}$$

$$V_{r(rms)} = \frac{V_{r(p-p)}}{2\sqrt{3}}$$

$$V_{dc} = V_m - \frac{V_{r(p-p)}}{2}$$

$$V_{dc} = V_m - \frac{I_{dc}}{2fC}$$

$$V_{dc} = V_m - \frac{I_{dc}}{4fC}$$

$$V_{r(rms)} = \frac{I_{dc}}{2\sqrt{3}fC} = \frac{V_{dc}}{2\sqrt{3}fCR_L}$$

$$V_{r(rms)} = \frac{I_{dc}}{4\sqrt{3}fC} = \frac{V_{dc}}{4\sqrt{3}fCR_L}$$

$$r = \frac{I_{dc}}{2\sqrt{3}fCV_{dc}} = \frac{1}{2\sqrt{3}fCR_L}$$

$$r = \frac{I_{dc}}{4\sqrt{3}fCV_{dc}} = \frac{1}{4\sqrt{3}fCR_L}$$

$$V'_{dc} = \frac{R_L}{R + R_L}.V_{dc}$$

$$V'_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X_C^2}}.V_{r(rms)}$$

$$X_C = \frac{1}{2\pi fC} \qquad X_C = \frac{1}{4\pi fC} \qquad r' = \frac{V'_{r(rms)}}{V'_{dc}}$$

$$V'_{r(rms)} = \frac{X_C}{R}.V_{r(rms)}$$

$$r' = rX_C\left(\frac{R + R_L}{R.R_L}\right)$$

$$V'_{dc} = V_{dc} - I_{dc}R_1$$

$$V'_{dc} = \frac{R_L}{R_L + R_1}.V_{dc}$$

$$V'_{r(rms)} = \frac{V_{r(rms)}}{(2\pi f)^2 LC}$$

$$V'_{r(rms)} = \frac{V_{r(rm)}}{(4\pi f)^2 LC}$$

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$$VR = \frac{V_{NL} - V_{FL}}{V_{FL}}$$

$$\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$$

$$2V_m = V_{c2} = V_m + V_{c1}$$

$$3V_m = V_{c1} + V_{c3} = V_m + 2V_m$$

$$S = \frac{\Delta V_o}{\Delta V_i}$$

$$V_R = V_i - V_z$$

$$R_{s(\min)} = \frac{V_{i(\max)} - V_z}{I_{z(\max)}}$$

$$I_z = \frac{P_z}{V_z}$$

$$R_{L(\min)} = \frac{V_Z}{V_{i(\max)} - V_Z} . R_S$$

$$V_o = V_r - V_{be}$$

$$R_c = \frac{V_{cc} - V_{ce}}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be}}{I_b}$$

$$\beta = \frac{I_c}{I_b}$$

$$C_e \geq \frac{10}{2\pi f R_e}$$

$$V_e = \frac{V_{cc}}{10}$$

$$R_e = \frac{V_e}{I_e} \simeq \frac{V_e}{I_c}$$

$$R_c = \frac{V_{cc} - V_{ce} - V_e}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be} - V_e}{I_b}$$

$$R_{b1} = \frac{R_{b2}(V_{cc} - V_b)}{V_b}$$

$$R_{b2} = \frac{1}{10}\beta R_e$$

$$V_b = V_e + V_{be}$$

$$V_{be} = h_{ie} i_b + h_{re} V_{ce}$$

$$i_c = i_{fe} i_b + h_{oe} V_{ce}$$

$$A_i = \frac{h_{fe}}{1 + h_{oe} Z_L}$$

$$A_i = h_{fe}$$

$$A_i = \left(\frac{h_{fe}}{1 + h_{oe} Z_L}\right)\left(\frac{R_b T}{R_{bT} + Z_1}\right)\left(\frac{R_c}{R_c + R_L}\right)$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie} + (h_{ie} h_{oe} - h_{fe} h_{re}) Z_L}$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie}}$$

$$Z_1 = h_{ie} - \frac{h_{fe} h_{re} Z_L}{1 + h_{oe} Z_L}$$

$$Z_1 = h_{ie}$$

$$Z_2 = \frac{1}{h_{oe} - \frac{h_{fe} h_{re}}{h_{ie} + R_s}}$$

$$Z_2 = \frac{1}{h_{oe}}$$

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$$A_p = \frac{A_i^2 R_L}{R_1} = -A_v A_i$$

$$A_p = \frac{h_{fe}{}^2 R_L}{h_{ie}}$$

$$Z_0 = R_C // R_L // Z_2 = Z_L // Z_2$$

$$Z_0 = R_C // Z_2 = Z_L // Z_2$$

$$Z_{IN} = R_B // Z_1$$

$$Z_i = R_{b1} // R_{b2} // Z_1$$

$$I_1 = \frac{R_{bT} I_i}{R_{bT} = Z_1}$$

$$I_0 = h_{fe} I_b = h_{fe} \left(\frac{R_{b2}(I_i)}{R_{b2} + h_{ie}} \right)$$

$$A_i = \frac{I_0}{I_1}$$

For common base, substitute all the 'e' subscripts with a 'b' in the h-parameters.

$$Z_L = R_c // R_L$$

$$I_1 = \frac{R_e I_i}{R_e + Z_1}$$

$$CMRR = \frac{A_{dm}}{A_{cm}}$$

$$CMRR\ (dB) = 20log \frac{A_{dm}}{A_{cm}}$$

$$I_e = \frac{V_e}{R_e}$$

$$I_c = \frac{I_e}{2}$$

$$R_L = \frac{V_{R_L}}{I_C}$$

$$g_m R_L = \frac{h_{fe}}{h_{ie}} . R_L$$

$$V_0 = -\left(\frac{R_f}{R_1} \right) . V_i$$

$$V_0 = \left(\frac{R_f}{R_1} + 1 \right) . V_i$$

$$V_0 = -\left(\frac{R_f}{R_1} . V_1 + \frac{R_f}{R_2} . V_2 + \frac{R_f}{R_3} . V_3 \right)$$

$$V_0 = -\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) R_f$$

$$V_0 = -(V_1 + V_2 + V_3)$$

$$V_0 = -(I_1 + I_2 + I_3)\ R_f$$

$$V_0(t) = -\frac{1}{RC} \int V_i(t)$$

$$V_0(t_b) = -\frac{1}{RC} \int_{t_a}^{t_b} V_i(t_b) + V_c(t_a)$$

$$t = \frac{1}{f}$$

$$A_v = -\frac{R_s}{R_1}$$

$$R_2 = \frac{R_1 R_s}{R_1 + R_s}$$

$$f_c = \frac{1}{2\pi R_s C}$$

$$V_0(t) = -RC \frac{dV_i(t)}{dt}$$

$$A = \frac{R_f}{X_c}$$

$$A = -\frac{R_f}{R_s}$$

$$V_0(t) = -R_f C \frac{d}{dt} . v_i \sin \omega t$$

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$t = R_f C$

$V_0 = A(V_r - V_i)$

$V_0 = \frac{R_f}{R_s}(V_2 - V_1)$

$V_0 = V_2 - V_1$

$f_0 = \frac{1}{2\pi\sqrt{C_1 C_2 R_1 R_2}}$

$f_0 = \frac{1}{2\pi RC}$

$f_0 = \frac{1}{2\pi\sqrt{L_T C_1}}$

$L_T = L_1 + L_2 + 2M$

$f_0 = \frac{1}{2\pi\sqrt{LC_T}}$

$C_T = \frac{C_1 C_2}{C_1 + C_2}$

$f_0 = \frac{1}{2\pi\sqrt{LC_2}}$

$f = \frac{1}{2\pi RC\sqrt{6}}$

$f_0 = \frac{1,5}{RC}$

$f_0 = \frac{1}{t} = \frac{1}{t_1 + t_2}$

$t_1 = 0,7\ R_2 C_1$

$t_2 = 0,7\ R_1 C_2$

$f_0 = \frac{1}{1,4RC}$

$V_i = I_{c2} R_e + V_{be(ON)}$

$t = 1,1\ RC$

$f_0 = \frac{1,443}{(R_A + 2R_B)C}$

$t_{low} = 0,693\,(R_B)C$

$t_{high} = 0,693\,(R_A + R_B)C$

$t_T = t_{low} + t_{high}$

$K = \frac{\Delta R / R}{\Delta l / l}$

$\sigma = \Delta l / l$

$R = \rho \frac{1}{\pi d^2 / 4}$

$\sigma = \frac{S}{E}$

$Resolution = \frac{1}{number\ of\ turns}$

$Resolution = \frac{voltage\ drop\ across\ adjacent\ turns}{total\ voltage\ drop}$

$R_t = Ae^{B/T}$

$T = 273 + °C$

$V_A = \frac{R_2}{R_1 + R_2}.V_T$

$V_B = \frac{R_t}{R_t + R_3}.V_T$

$V_{AB} = V_A - V_B$

$A_v = \frac{V_0}{V_i}$

$V_{Hall} = kIH$

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